

ISMAILIA, Egypt

WMO: 624400

Lat: 30.592N

Lon: 32.247E

Elev: 41

StdP: 14.67

Time Zone: 2.00 (E02)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.2	46.2	31.3	25.6	62.3	34.0	28.7	61.5	28.2	58.8	24.3	60.9	2.9	270	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.7	99.9	71.8	97.6	72.0	95.5	72.0	77.9	88.0	76.8	87.2	75.9	86.2	11.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	133.3	81.7	74.1	127.4	80.8	73.2	123.6	80.3	41.4	88.1	40.3	86.8	39.4	86.2	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.5	19.5	17.8	DB	40.6	106.6	2.4	2.9	38.9	108.7	37.4	110.3	36.1	112.0	34.3	114.0	
			WB	38.0	80.6	2.1	2.1	36.5	82.1	35.2	83.4	34.0	84.5	32.5	86.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	71.8	57.4	59.2	63.4	69.7	76.2	81.1	84.6	85.0	81.3	75.5	67.4	60.1
	DBStd	10.62	3.62	4.65	5.75	5.93	4.53	3.68	2.54	2.30	3.02	4.20	4.36	3.63
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD65	703	237	173	101	13	0	0	0	0	0	0	22	157
	CDD50	7967	231	258	416	592	811	934	1074	1085	940	791	522	313
	CDD65	3193	2	11	51	155	346	484	609	620	490	326	94	5
	CDH74	33819	24	116	480	1470	3318	5299	7455	7514	5030	2518	548	47
	CDH80	16266	1	32	166	615	1494	2721	4008	3919	2339	862	105	4
(14) Wind	WSAvg	8.0	7.0	8.3	8.8	9.2	8.8	8.7	8.6	8.2	7.9	7.1	6.2	6.7
(15) Precipitation	PrecAvg	0.9	0.2	0.2	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
	PrecMax	1.8	0.8	0.5	0.8	0.9	0.2	0.0	0.0	0.0	0.0	0.2	0.6	0.5
	PrecMin	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.5	0.3	0.2	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.2	84.5	92.4	99.4	100.8	102.5	102.3	100.6	101.3	96.9	87.8	79.2
		MCWB	59.1	60.7	63.3	66.3	68.1	71.6	73.0	73.9	70.9	68.9	66.1	60.0
	2%	DB	71.8	76.8	84.3	92.8	95.1	98.3	98.8	98.6	95.8	90.0	82.6	73.9
		MCWB	56.8	57.3	61.2	64.2	66.2	70.9	73.2	74.2	71.9	69.7	65.5	59.1
	5%	DB	68.4	71.9	78.9	87.0	91.5	94.9	96.7	96.6	92.9	87.2	79.4	71.4
		MCWB	56.1	56.3	60.4	62.4	66.1	70.4	73.0	73.8	72.0	69.6	64.4	58.4
	10%	DB	66.2	68.7	74.4	82.1	88.0	92.2	94.8	94.3	90.2	84.5	76.9	69.2
		MCWB	55.1	55.6	58.7	61.6	65.9	70.1	73.2	73.7	71.8	69.0	63.7	57.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.8	63.3	67.5	69.3	71.1	76.2	78.9	79.9	77.3	75.5	70.2	63.7
		MCDB	71.7	78.5	82.6	90.6	88.1	88.8	89.4	89.5	86.5	84.9	79.5	71.7
	2%	WB	59.0	59.8	63.6	66.2	69.5	74.4	77.3	78.5	75.7	73.2	68.3	61.7
		MCDB	68.3	71.3	79.4	84.8	85.7	87.6	88.3	88.0	85.7	81.8	77.6	70.6
	5%	WB	57.4	58.1	61.5	64.4	68.4	73.3	76.4	77.3	74.6	71.9	66.9	60.1
		MCDB	66.3	68.2	74.5	81.2	83.2	86.3	87.3	87.4	84.8	81.5	75.6	68.9
	10%	WB	56.1	56.6	59.8	62.9	67.4	72.3	75.5	76.3	73.7	70.7	65.3	58.7
		MCDB	64.5	66.3	71.4	77.4	81.6	85.0	86.2	86.4	83.9	80.5	73.9	67.0
(35) Mean Daily Temperature Range	5% DB	MDBR	18.5	18.7	20.2	22.5	23.5	22.9	21.6	20.7	20.2	19.5	19.9	18.9
		MCDBR	21.7	24.1	26.4	30.0	28.4	26.4	23.9	22.7	22.5	21.6	21.8	21.8
	5% WB	MCWBR	10.9	9.8	9.2	9.3	7.5	6.4	5.9	6.0	5.9	6.4	8.3	10.4
		MCWBR	20.1	20.8	24.0	25.7	24.6	24.0	22.2	21.4	21.1	20.4	20.5	20.2
(40) Clear-Sky Solar Irradiance	taub		0.402	0.441	0.487	0.534	0.536	0.456	0.451	0.454	0.459	0.468	0.410	0.395
		taud	2.084	1.964	1.852	1.764	1.787	2.027	2.057	2.052	2.029	1.998	2.135	2.131
	Ebn at Noon		254	255	252	246	246	266	266	264	258	244	250	249
		Edh at Noon	43	53	63	71	70	55	53	53	52	50	40	39
(44) All-Sky Solar Radiation	RadAvg		1096	1364	1786	2147	2416	2589	2530	2333	2005	1575	1207	1018
	RadStd		34	62	70	77	69	31	27	27	26	42	35	30

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+1.08	N/A	-2.45	N/A	N/A	N/A	N/A	-90	+382	+308

Nomenclature: See separate page